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Wireless Driving of Electrostatic Film Actuator by Pre-charging Electrodes with DC High Voltages

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Abstract. Electrostatic film actuator is a thin, lightweight and flexible actuator composed of two plastic films with fine-pitched electrodes embedded in them. In this paper, we report a novel driving method for the actuator, in which the electrodes in one film (the slider film) are pre-charged with DC high voltages, and the slider film is driven by AC voltages applied to the electrodes in the other film (the stator film). Typically, the actuator has been driven by applying three-phase AC voltages to the electrodes in both films; in the driving method, the slider film required connections of feeding wires, whose tensions causing disturbance to its motion. The advantage of the pre-charging method is that the slider film is free from the feeding wires; in turn, it requires an additional step before driving: charging of the slider electrodes. To confirm the feasibility of the method, experiments were conducted using a pair of films fabricated by flexible printed circuit technology. Each of the films incorporate 475 strips of electrodes within their overlapping area; the electrodes are 18 mm in length and 40 μm in width, arranged at a constant pitch of 160 μm. DC voltages used for pre-charging the three-phase electrodes in the slider were 500 V, -500 V and 0 V, respectively. After the pre-charging, 500 V three-phase voltage applied to the stator drove the slider with maximum thrust force of approximately 120 mN. We also found that the thrust force of the actuator decay with time after pre-charging, which indicate that further investigations are necessary to prolong its duration.

Introduction

An electrostatic film actuator is a linear surface actuator composed of a pair of thin and flexible plastic films with patterned strip electrodes [1]. The actuator is a synchronous actuator, driven at a speed proportional to the frequency of the voltages applied to the electrodes. To extend the actuator's range of application, the authors have recently developed a 2-DOF planar electrostatic actuator [2]. As the actuator gained another DOF, the actuator requiring feeding wires to the slider (the moving film) arose to be a problem; construction of cable carriers with little disturbance is always a problem in planar actuators. In this paper, we report a driving method for the actuator that frees the slider from feeding wires. The method involves charging of electrodes with dc high voltages before driving.

Various synchronous electrostatic actuators have been reported which operate without direct feeding of power to its moving part. The first type of actuator is a variable capacitance motor with salient poles, actively studied as micromotors in MEMS, both in rotary and linear forms [3, 4]. The second type of actuator is an electret motor, which utilizes electrets: dielectrics that have quasi-permanent charges [5, 6]. Although the principles of these two actuators may lead to a realization of film actuators with wireless slider, their implementations in flexible films are not established. The third actuator is a film actuator with large-area electrodes that electrostatically induce voltages on fine-pitched electrodes [7]. The actuator realizes wireless slider while maintaining thin film structure. However, its application to 2-DOF planar actuator is difficult since the large-area electrodes require position independent overlapping area. For these reasons, we started looking for an alternative way to realize a synchronous electrostatic film actuator with wireless slider, which is extendable to 2-DOF actuator.

The driving method we propose in this paper comes from an observation of a phenomenon that has long been known in our research group: the slider of the actuator, after it has been unplugged from the

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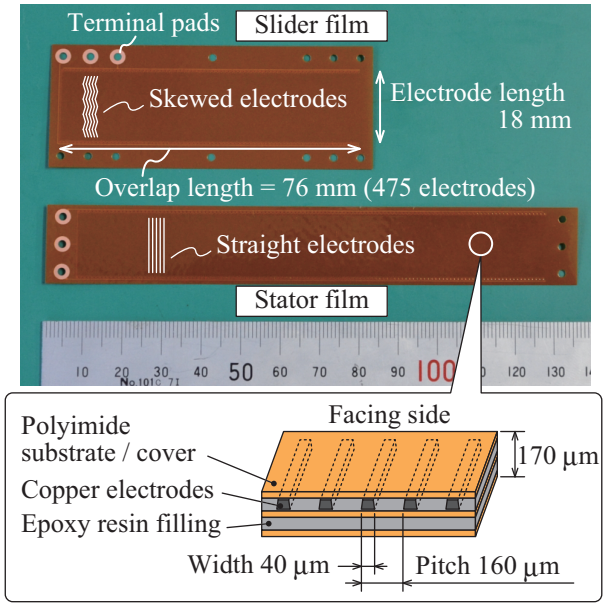


Fig. 1: Photograph of the stator and slider of the electrostatic film actuator.

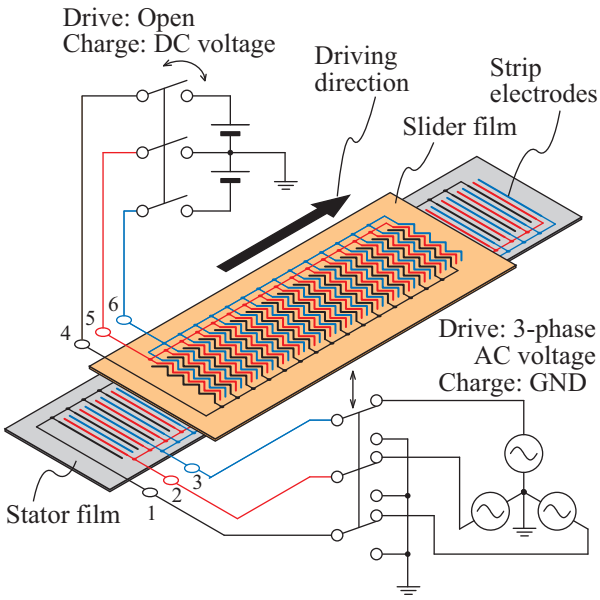


Fig. 2: Schematic of an electrostatic film actuator driven by DC pre-charging method.

voltage sources, keeps on moving for some time. In this paper, we clarify that the phenomon is caused by the charges and potentials remained on the slider electrodes, just like in a charged, open-circuited capacitor. As we excite the stator with AC voltages, electrostatic forces acting on the remained charges drive the unplugged slider. Unlike electret motors, charges decay quickly because of the leakage current. Thus, the driving method is imperfect with limited duration; however, it gives us an insight towards realization of continuous wireless driving.

Remainder of this paper is as follows. First, the procedure of the driving method is explained, together with details of the electrode structure. Second, a theoretical model is provided to explain the mechanism of the driving method. Then, we describe the results of experiments that verify the theory, which include observation of decay in thrust force and electrode potential. The results include one unexpected finding: when a corona discharge occurs during the charging process, the force appears to last longer. This phenomenon is unexplainable with our current theory, but is included in this paper to indicate a direction in future studies. Finally, the last section concludes the paper.

Structure of Film Actuator and Procedure of Pre-charged Driving

Figure 1 shows a photograph of the films used in this work. The two films: the stator film and the slider film are flexible printed circuits with three-phase electrodes patterned on their polyimide substrate. Stator electrodes are straight lines, 40 μm in width, 18 mm in length, and pitched at an equal distance of 160 μm. Slider electrodes are skewed by 1.5 times the pitch to reduce thrust force ripple as proposed in [8]. When driving, the slider film is placed on top of the stator film, with glass beads of 20 μm diameter scatterd between them to maintain an air gap.

Electrical connection for the proposed driving method is shown in Fig. 2, and the procedure for the pre-charged driving is as follows. First step is the charging of slider electrodes by applying DC voltages. During this step, the stator electrodes are grounded. In the next step: the driving step, the DC voltages are suddenly removed, and the slider electrodes are made open-circuited. At the same time, the stator electrodes are fed with three-phase AC voltage.

Principle of the pre-charged driving method is shown in Fig. 3. The DC voltages applied during charging step raise the potentials of the slider electrodes and form a periodic potential distribution on the slider surface. In driving step, a traveling potential wave is formed on the stator surface by the three-phase AC voltage. The three-phase voltage excites a traveling electric field in the gap between the films and the field exerts forces on the charges induced on the slider surface, resulting in giving

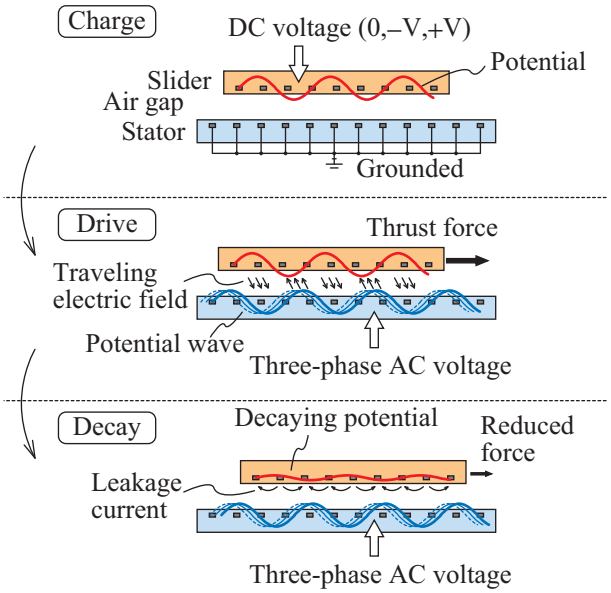


Fig. 3: Principle of DC pre-charged operation of electrostatic film actuator.

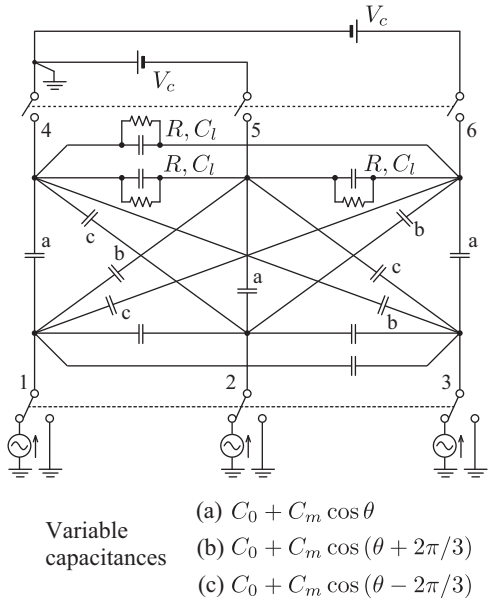


Fig. 4: Equivalent circuit model of electrostatic film actuator for transient analysis.

the slider a thrust. At the instant of switching, the slider potentials are at the same level as the charged voltage, but they decay exponentially with time because the insulation among electrodes is imperfect; leakage currents flow among the electrodes and their potentials eventually even out. Thus, the thrust force of the actuator also decays as the time elapses.

Modeling and Transient Analysis

To estimate the actuator's behavior during pre-charged driving, we constructed an equivalent circuit model shown in Fig. 4, which is based on a capacitance network model of the electrostatic film actuator reported in [9]. In this model, the terminals numbered 1, 2 and 3 represent the stator electrodes and those with numbers 4, 5 and 6 represent the slider electrodes. The capacitances formed between stator and slider electrodes vary with the slider position x. Thus, they are represented by variable capacitances that vary with an electrical angle $\theta = (2\pi/3p)x$, where p is the electrode pitch. Symbols C_0 and C_m are the offset and the amplitude of the variable capacitances, C_l is the capacitance between neighboring electrodes and R represents the insulation resistance.

To calculate the thrust force, we first need to know the transient response of the slider potentials after switching from the charging state to the driving state. As an initial condition of the slider potentials, we assume $[v_4\ v_5\ v_5]^T = [0\ -V_c\ V_c]^T$ at $t = 0$, where v_k is the potential of the electrode number k and V_c is the DC voltage for charging. As an excitation voltage to the stator, we assume a three-phase voltage with amplitude V_d and angular frequency ω: $[v_1\ v_2\ v_3]^T = V_d[\sin\omega t\ \ \sin(\omega t - 2\pi/3)\ \ \sin(\omega t + 2\pi/3)]^T$. Further, for simplicity, we assume a quasi-static condition where slider speed is negligibly slow ($\dot{\theta} \approx 0$). Based on these assumptions, the transient response of the slider potential is calculated as

$$\begin{bmatrix} v_4 \\ v_5 \\ v_6 \end{bmatrix} = \begin{bmatrix} 0 \\ -V_c e^{-t/T} \\ V_c e^{-t/T} \end{bmatrix} + KV_d \left(e^{-t/T} \begin{bmatrix} \sin(\theta - \alpha) \\ \sin(\theta - \alpha + 2\pi/3) \\ \sin(\theta - \alpha - 2\pi/3) \end{bmatrix} + \begin{bmatrix} \sin(\omega t - \theta + \alpha) \\ \sin(\omega t - \theta + \alpha - 2\pi/3) \\ \sin(\omega t - \theta + \alpha + 2\pi/3) \end{bmatrix} \right). \quad (1)$$

The most important parameter in this expression is the time constant T, which is defined as $T = (C_{sl} + C_l)R/3$, where C_{sl} is the self-capacitance of slider electrodes: sum of all capacitances connected to a slider node in Fig. 4. The coefficient K and the angle α are frequency dependent variables, respectively defined as

$$K = \frac{3}{2}\frac{C_m}{C_{sl} + C_l}\frac{\omega T}{\sqrt{1 + \omega^2 T^2}} \text{ and } \alpha = \tan^{-1}(1/\omega T) \quad (2)$$

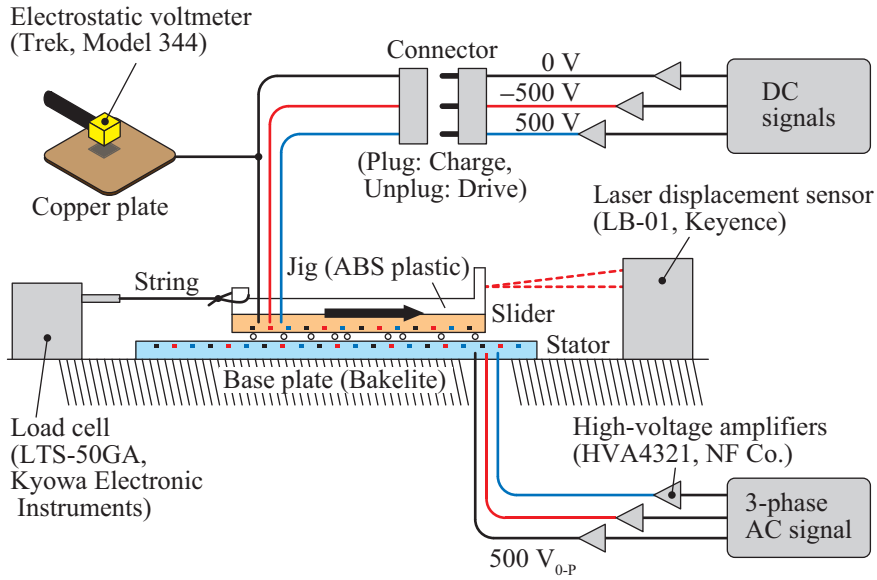


Fig. 5: Experimental setup for measuring decay of thrust force and electric potential on slider electrodes.

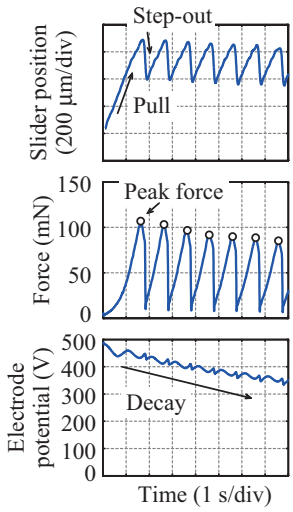


Fig. 6: Measured data in short time scale.

The first term in (1) is the decaying slider potential of this paper's concern. The second term is the voltages induced across the airgap due to the electrostatic coupling of stator and slider; first term in parenthesis is the transient, and the second term is the steady-state response. Such voltages play an important role in electrostatic induction motors [10, 11], but as will be shown in the experimental results, they are small enough to be neglected for the film actuator used in this work.

Now that the potentials are known, we can calculate the thrust force of the actuator by the virtual work principle, which yields

$$F = -\frac{\sqrt{3}\pi}{p} C_m V_d V_c e^{-t/T} \sin(\theta - \omega t) + \frac{3\pi}{2p} C_m K V_d^2 \left\{ -e^{-t/T} \sin(\omega t - 2\theta + \alpha) + \sin\alpha \right\}. \quad (3)$$

The first term in this expression is the force resulting from the pre-charged slider potential and the second term is the force brought by the voltages induced across the gap. Again, the former is of this work's main concern, and the latter is negligibly small. The force by pre-chaging indicates that the actuator is driven as a synchronous actuator: when the slider is working against a constant load, it travels at a constant velocity equal to $3p\omega/2\pi$.

Experimental Results

Experimental Setup and Conditions. To confirm the feasibility of the pre-charging method and to investigate on its duration, we conducted experiments using the setup shown in Fig. 5. The three important quantities to measure were the thrust force, the potentials on the slider electrodes, and the slider's position. To measure the force, we let the slider pull an elastic string tied to a load cell. As for the measurement of potentials, we measured the surface potential of a copper plate connected to the electrodes, using electrostatic voltmeter. Position of the slider in motion was recorded using a laser displacement sensor. For all experiments, charging voltage and the excitation voltage was set to (0, -500, 500) V_{DC} and 500 V_{0-P}, respectively. The actuator could operate at voltages 200 V and higher, but its force became unstable over 700 V due to corona discharge; thus, we chose 500 V to assure stable operation. The switching between charging and driving modes was realized by unplugging the connector feeding DC voltage to the slider. As for the length of charging time we chose 5 seconds. The charging time could be shorter, because the time it takes for the electrode potential to rise is very short, just like in charging a capacitor in parallel RC circuit. However, for the consistency of experimental conditions, we chose to maintain the charging time at 5 seconds.

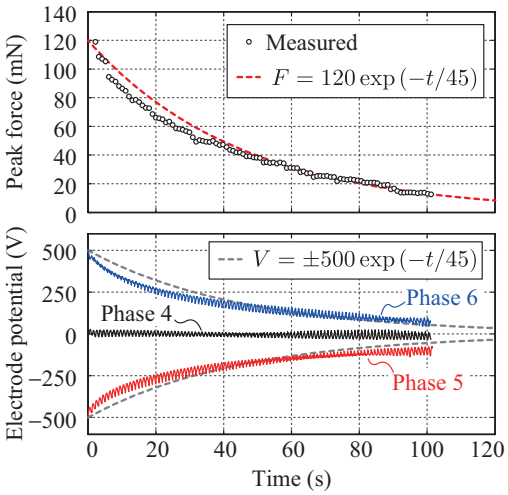
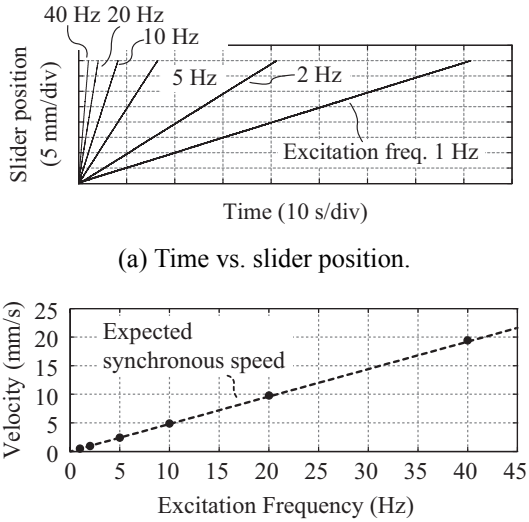


Fig. 7: Decay of thrust force and potential on slider electrodes, driven by 500 V_{0-P} / 1 Hz three-phase voltage.



(a) Time vs. slider position.

(b) Excitation frequency vs. slider velocity

Fig. 8: Motion and speed of the slider with no load, driven by 500 V_{0-P} three-phase voltage.

Observation of Decay. Figure 6 shows the measured slider position, thrust force, and potential on one of the slider electrodes. The excitation frequency is set at 1 Hz; thus, the electrostatic force act on the slider to synchronize its speed at 0.48 mm/s. However, as the tension of the string increases and reaches the maximum thrust force, the slider cannot withstand the tension and steps backwards. Thus, the peak force at stepping back corresponds to the maximum thrust force, and this force decreases with the decay of the slider potential. In the measured slider potential, an oscillation is observed with the same period as the slider's motion. This oscillation corresponds to the second term in (1), and it is small enough in comparison with the decaying trend. Figure 7 shows the same result plotted in longer time range. From this data, we can clearly observe the decay of potentials and the decrease of force, and the time constant is roughly approximated as T=45 s. The slider stopped 100 s after charging, when the thrust force could no longer overcome the friction between the films. Within this limited time, the slider could be driven at various speeds. Figure 8(a) shows the motion of the slider driven at different frequencies just after charging. In these experiments, the string was untied from the slider and full stroke travel of 40 mm was enabled. As plotted in Fig. 8(b), the velocity of the slider was proportional to the excitation frequency, which indicates that the actuator is driven synchronously.

Although the results show that the pre-charged driving method is feasible, it also indicates problems of this method. First, increasing of time constant is difficult, as it is determined as the product of electrode's capacitance and insulation resistance, which are difficult to be controlled separately. Insulating materials with higher permittivity and less leakage are to be considered. Second, because the length of duration relies on the insulation resistance, it may vary due to the condition of the films such as the amount of moisture content.

Long-lasting Force after Corona Discharge. During the experiments, we unexpectedly found that the thrust force after charging lasts much longer when the charging involves corona discharge in the air gap. In this case, unlike corona motors, which utilize continuous corona discharge to charge its rotor [12], the discharge is intentionally caused only before driving as a preparation. Figure 9 shows the force measured after this preparation. The discharge was triggered by applying high voltage of 1000 V_{0-P} / 1 Hz to the stator electrodes while the

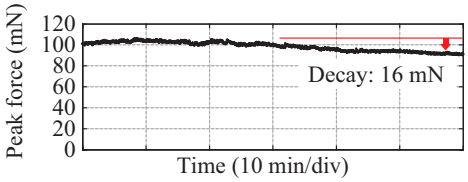


Fig. 9: Decay of thrust force after intentional corona discharge caused by stator voltage 1000 V_{0-P} / 1 Hz and slider voltage (0, -500, 500) V, driven at 500 V_{0-P} / 1 Hz.

slider is being charged with DC voltages. At this condition, we could observe spikes in currents fed from the source, which indicate the presense of discharge. Another point to mention is that the slider electrodes were once grounded before driving; thus, this phenomenon involves no rising or decaying of slider potentials. Eventhough we used the same set of films as in the previous section, the slider kept on pulling the string for more than an hour, and the decay observed during was only 16 mN, less than one fifth of the initial force. This phenomenon is unexplainable with the model explained in this paper, but we speculate that at this condition, charges were added to the slider surface to form electrets.

Conclusions

A novel driving method for electrostatic film actuators by pre-charging the electrodes with DC high voltages before driving was proposed in this paper. Behavior of the actuator in the driving method was explained and analyzed using a lumped-parameter model. Although the feasibility of the method was demonstrated through experiments, it also revealed that the duration of operation is in the order of minutes and prolonging of which appears to be challenging. We hope to investigate further on this method and the charging method involving corona discharge found as a by-product. Application of these methods to 2-DOF planar film actuator is also our future work.

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Precision Engineering and Nanotechnology V
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